

Experiment Title

Basic VLAN Configuration and Verification Using Cisco Switch CLI

Objectives

The objective of this experiment is to understand the concept of Virtual Local Area Networks (VLANs) and their role in logically segmenting a network. This experiment focuses on configuring VLANs on a Cisco switch using the Command Line Interface (CLI) and verifying communication between devices belonging to the same VLAN while ensuring isolation between different VLANs.

System Configuration

Software Platform:

Cisco Packet Tracer (Version 8.x or above)

Operating System:

Windows 11

Network Components:

- One Cisco 2960 Switch
- Four Personal Computers (End Devices)

Cabling Used:

Copper Straight-Through cables (PC to Switch)

Configuration Method:

Cisco Switch Command Line Interface (CLI)

Steps and procedures

Step 1: Placing Devices on Cisco Packet Tracer Workspace

At the beginning of the experiment, a new project was created in Cisco Packet Tracer. One switch and four PCs were placed on the workspace. The devices were arranged properly to represent a small switched network and renamed where necessary for easier identification during configuration.

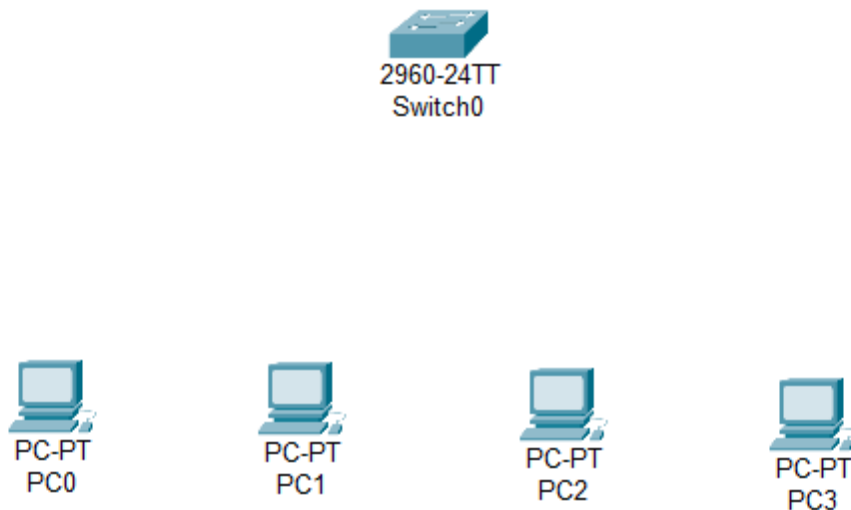


Figure 10.1: Placing all the devices into workspace

Step 2: Connecting Network Devices

Each PC was connected to the switch using copper straight-through cables. Different FastEthernet ports on the switch were used for each PC to allow individual port-based VLAN assignment later in the experiment.

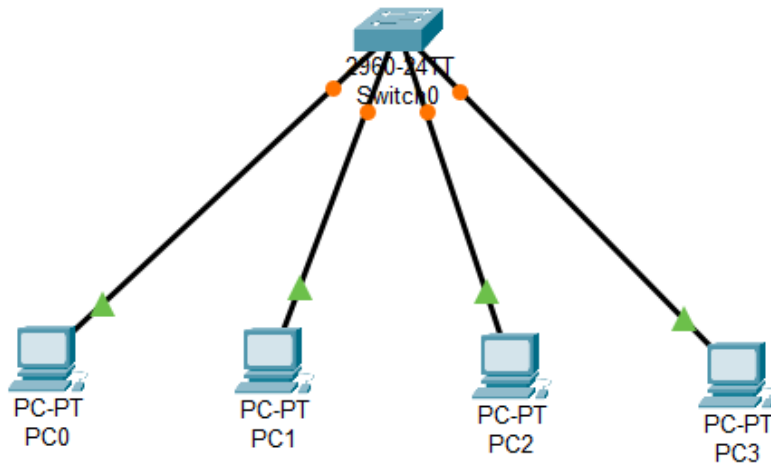


Figure 10.2: Connecting all devices through straight through cable

Step 3: Assigning IP Addresses to all PCs

Static IP addresses were assigned to all PCs. The devices were divided into two separate IP networks so that VLAN-based communication behavior could be observed clearly during testing.

PC	IP Address	Subnet Mask
PC0	192.168.1.2	255.255.255.0
PC1	192.168.1.3	255.255.255.0
PC2	192.168.3.2	255.255.255.0
PC3	192.168.3.4	255.255.255.0

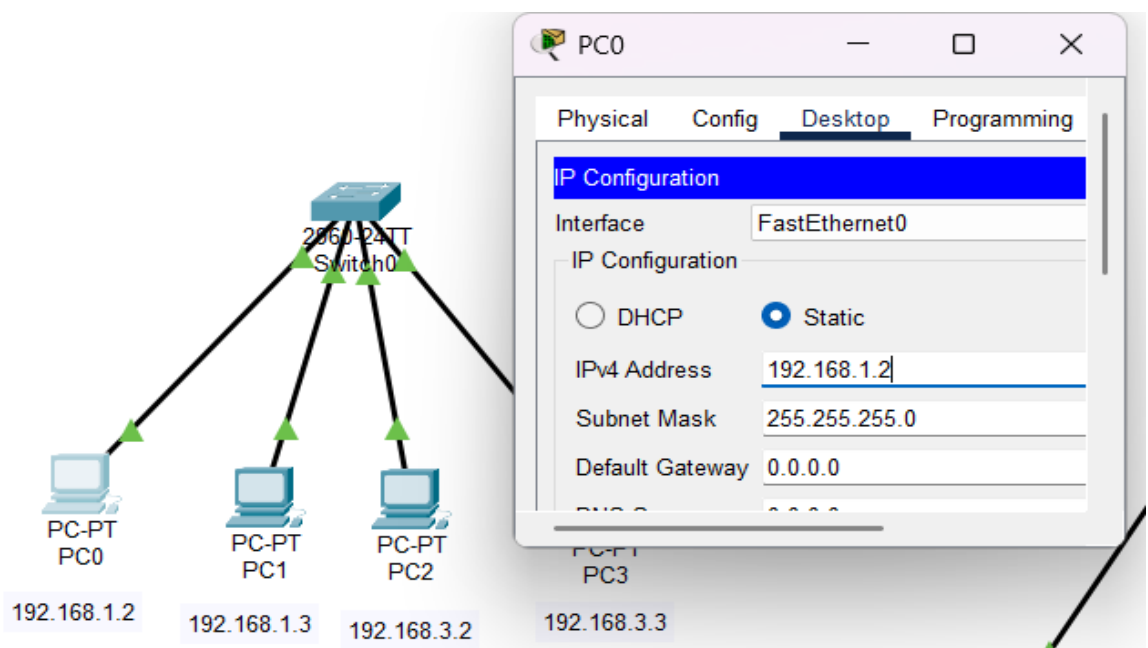


Figure 10.3: Assigning IP Addresses to all PCs

Step 4: Accessing the Switch CLI

The switch's Command Line Interface was accessed to perform VLAN configuration tasks. Privileged EXEC mode and global configuration mode were entered to allow VLAN creation and port assignment.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>enable
Translating "enable"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch>enable
Switch#
```

Figure 10.4: Accessing the Switch CLI

Step 5: Creating VLANs

Two VLANs were created on the switch using CLI commands. Each VLAN was assigned a unique VLAN ID and an appropriate name to distinguish between different logical groups within the network.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Accounts
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Figure 10.4: Creating VLANs using CLI

Step 6: Assigning Switch Ports to VLANs

Specific FastEthernet ports on the switch were configured as access ports and assigned to the corresponding VLANs. This step ensured that devices connected to those ports became members of their designated VLANs.

```

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fas
Switch(config)#interface range fastEthernet 0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fas
Switch(config)#interface range fastEthernet 0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switch access vlan 20
Switch(config-if-range)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#

```

Figure 10.5: Assigning Switch Ports to VLANs

Step 7: Verifying VLAN Configuration

To confirm successful VLAN creation and port assignment, verification commands were executed on the switch. The VLAN summary output displayed all configured VLANs along with their associated switch ports.

```

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	sales	active	Fa0/1, Fa0/2
20	Accounts	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#

```

Figure 10.6: VLAN Verification Output

Step 8: Testing VLAN Connectivity

Connectivity tests were performed using the ping command from the PCs. Devices within the same VLAN were able to communicate successfully, while communication attempts between devices in different VLANs failed. This behavior confirmed proper VLAN isolation.

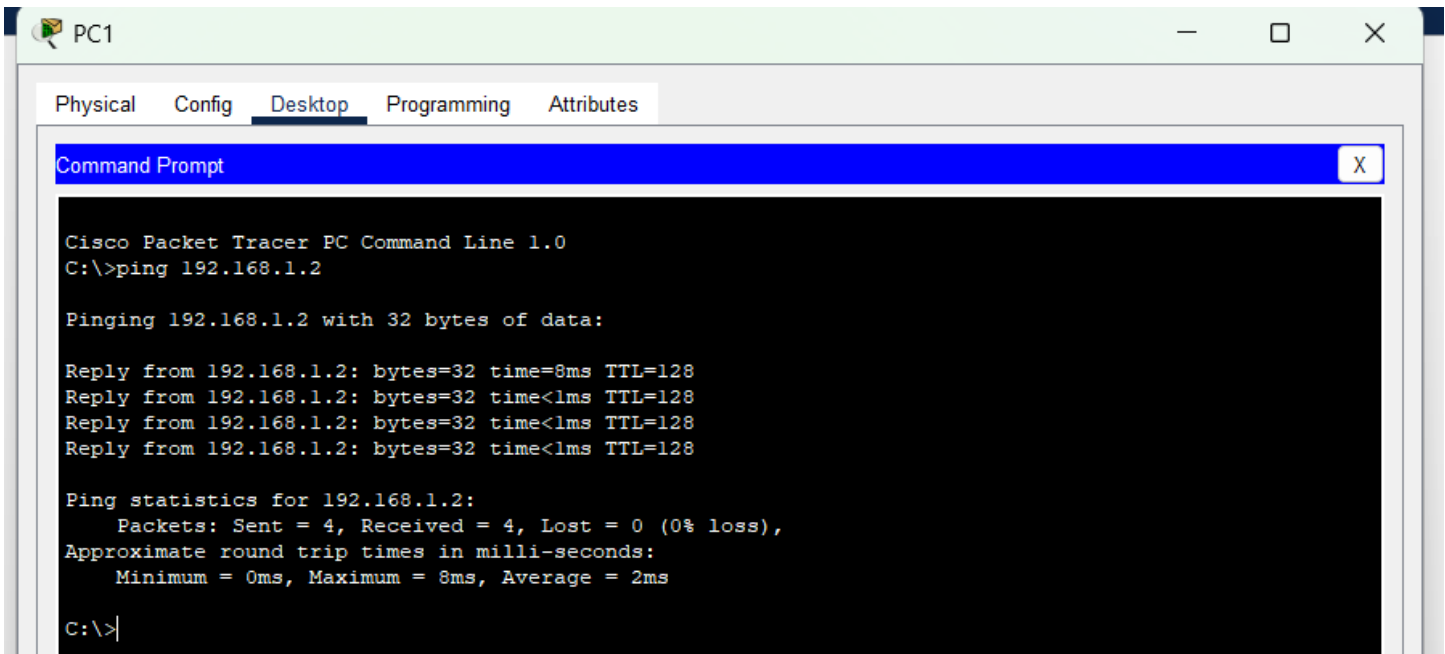


Figure 10.7: VLAN Communication Test Results

Step 9: Saving the Configuration

Finally, the switch configuration was saved to non-volatile memory to ensure that VLAN settings would remain intact after a device restart.

```
Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

Figure 10.8: Saving VLAN Configuration to NVRAM

Result and Discussion

The experiment successfully demonstrated VLAN configuration using a Cisco switch. VLANs were created and switch ports were correctly assigned, resulting in proper logical segmentation of the network. Devices belonging to the same VLAN communicated without issues, while devices in different VLANs remained isolated. This confirms the effectiveness of VLANs in improving network security, organization, and traffic management.

Conclusion

In this experiment, VLANs were configured using Cisco switch CLI commands to logically divide a single physical network into multiple virtual networks. The results verified that VLANs allow communication only within the same logical group while preventing unnecessary traffic between different groups. This experiment provided a clear practical understanding of VLAN concepts and highlighted their importance in modern switched networks.

